Dental Cements part 2

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Glass Ionomer Cement

- Glass ionomer has a wide range of uses in dentistry.
- Water based cement.

<u>Mode of supply:</u>

1. Powder and liquid (traditional form).
2. Water settable cement (the acid is freeze dried & added to the glass powder. the liquid may be distilled water or dil. Sol. Of tartaric acid).
3. Pre-proportioned capsule (for mechanical mixing).
4. Paste-paste system.

Glass Ionomer Cement

<u>**Mode of supply:**</u>

3. ***pre-proportioned capsules:***

Advantages:

a) It is loaded with the correct P/L.

b) Mechanical mixing (no hand mixing, save time).

c) Dispensing the material through the gun allows better adaptability to the prepared tooth.

Disadvantages:

High cost.

Glass Ionomer Cement

<u>Mode of supply:</u>

4- paste-paste system

- The material is supplied in a double barrel syringe (each barrel contains a paste).
- Proper proportioning and ease of use .

Disadvantages :

High cost.

Glass Ionomer Cement

Composition:

1. powder:

- Calcium- flouro-alumino-silicate powder(max. particle size 15µm).
- It consists of three main components silica(SiO_2) ,alumina (Al_2O_3), mixed with flux of sodium and calcium flouride.
- Barium glass is added to powder to give radiopacity.

2. liquid:

Homopolymer of acrylic acid OR Copolymer of acrylic acid, itaconic acid and maliec acid (to decrease viscosity of the liquid).

Tartaric acid to increase working time, decrease setting time.

Glass Ionomer Cement

Classification

- The most practical and widely used classification of GICs is based on their clinical application.
- According to this classification, there are three main types of GICs:

Type I: GICs used for luting applications.

Type II: GICs used as restorative materials.

Type III: GICs used as liners or bases.

Glass Ionomer Cement

Setting reaction

Acid-base reaction .

- Once the powder and liquid are mixed together, the reaction goes through the following three stages:

1. Dissolution
2. Migration
3. Reaction And Precipitation

Glass Ionomer Cement

<u>Setting reaction:</u>

1.Dissolution:

- Dissolution of the surface of glass by acid.
- H^+ attacks the powder particles and dissolve the outer layer (20% to 30 % of the particles are dissolved).
- Na^+, Ca^{++} ,Al^{+++} and F^- ions are released while the SiO_2 remains on the outer layer of the particles.
- $Ca^{++}$and Al^{+++} are responsible for cross linking the acid chains while Na^+ and F^- are released from the cement as NaF

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Setting reaction

2. Migration:

- Migration of the released surface ions (Ca^{++} ,Al^{+++} and F^-)into the liquid.
- The divalent calcium ions migrate first followed by trivalent Al ions.
- The sodium ions form silica gel on the surface of the glass particles.

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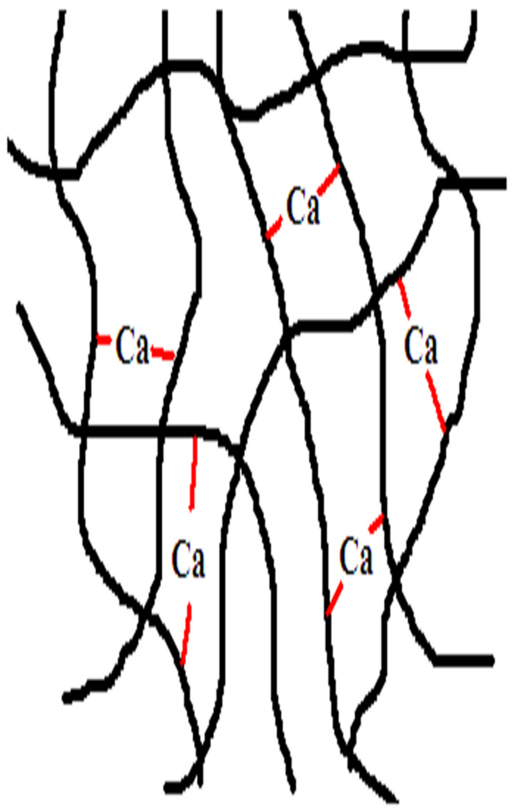
Setting reaction:

3. Reaction and precipitation:

Ca++

- The migrating Ca ++ ions react first with the carboxylic groups of the acid to form a cross-linked polycarboxylic salt gel.
- Each Ca ++ ion is divalent and can only cross-link two polycarboxylic chains.
- This results in a relatively weak cross-linked gel.
- This early Ca-mediated cross linking only leads to an “initial set” of the glass ionomer.



Initial setting
Ca bridges only

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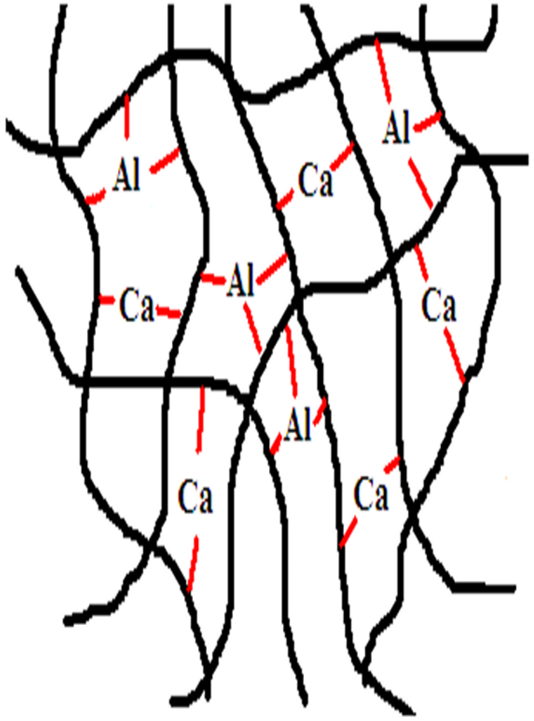
Setting reaction:

3. Reaction and precipitation:

Al+++ :

- followed by the reaction of the slowly migrating trivalent Al +++ ions.
- Being trivalent, each Al +++ ion can cross-link three polycarboxylic chains → more effective cross linking and formation of a stronger gel.
- This last reaction takes longer time (up to 24 hours) and results in the "final set" of the cement



Final setting
Ca and Al bridges

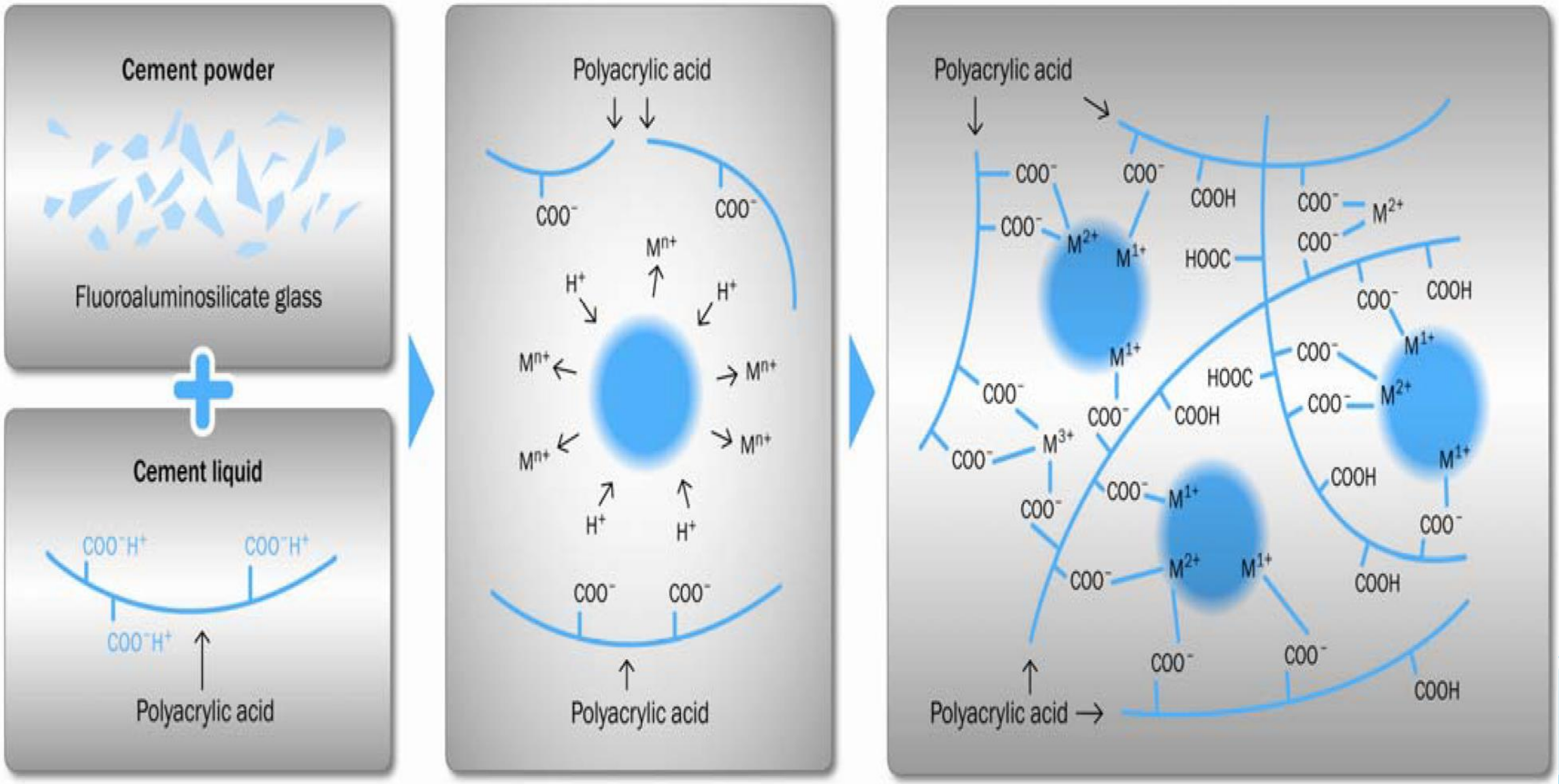
Glass Ionomer Cement

Setting reaction:

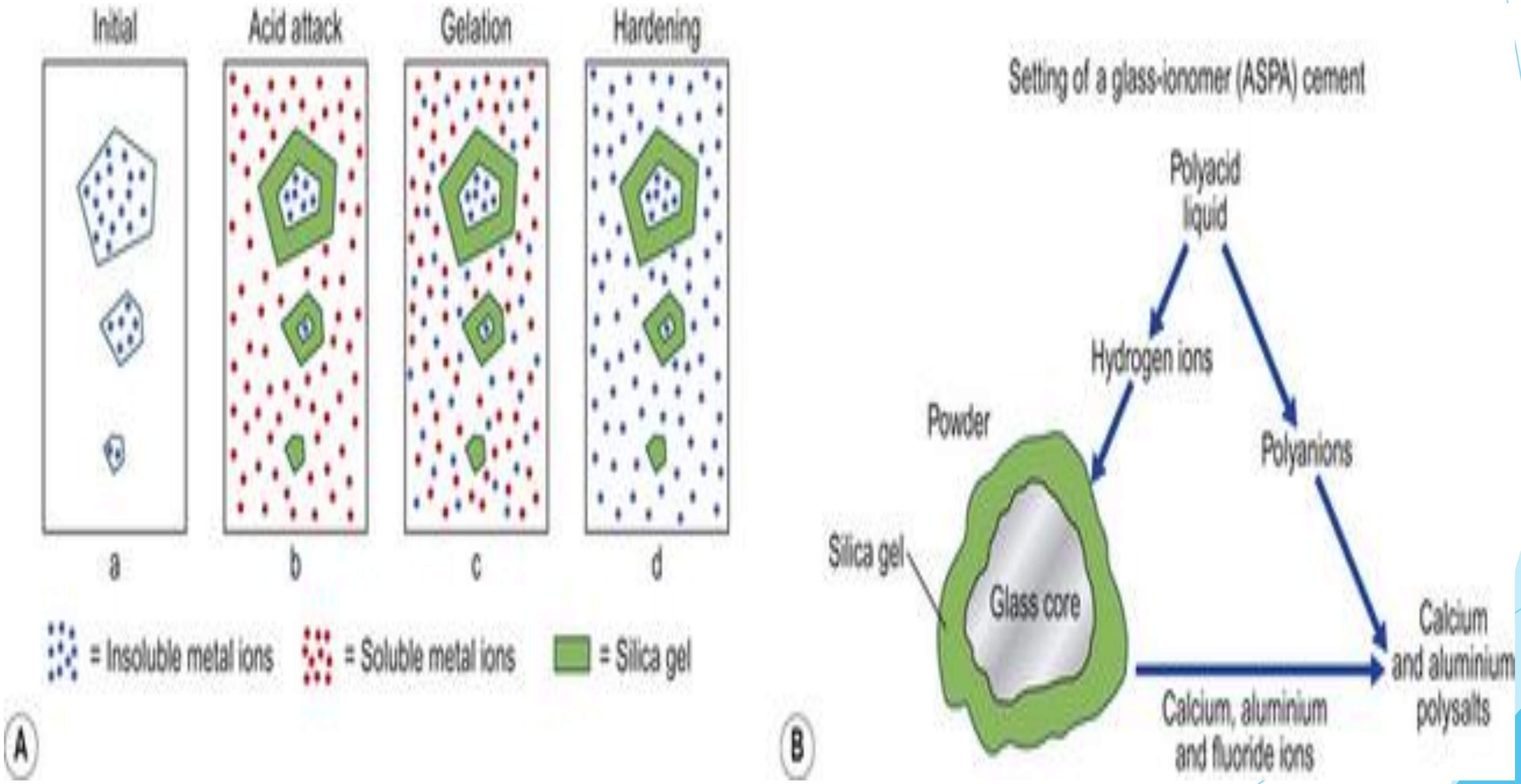
3. Reaction and precipitation:

- The water is bonded to silica gel (SiO_2) at the outer layer of the particles.
- This reaction takes long time (up to 24 h).
- Therefore ,the cement should be protected against premature exposure to saliva why?? as it affects the setting and the surface hardness of the cement.

Glass Ionomer Cement



Glass Ionomer Cement



Glass Ionomer Cement

Role of water in the setting process:

- Water is an important constituent of the cement liquid.
- It serves initially as a reaction medium and is loosely bound to the structure then,
- It slowly hydrates the cross-linked matrix and becomes tightly bound, thereby,
- Increasing the material strength by producing a stable gel structure.

Glass Ionomer Cement

Water issues during initial setting of GI

Dryness

The reaction will not go to completion, and the surface will crack.

Moisture

Dissolution of the matrix and ions will occur.

Both conditions result in a glass ionomer cement with reduced strength, increased solubility and poor aesthetics.

Glass Ionomer Cement

- Therefore, the glass ionomer cement must be protected against water changes (dryness or moisture contamination) in the structure during the setting process.

by covering the freshly placed glass ionomer cement by a varnish. Why??

Glass Ionomer Cement

Manipulation:

- P/L differs according to purpose:
- ↑ P/L for restoration
- ↓ P/L for luting

Glass Ionomer Cement

Manipulation:

- Mixing is done using plastic spatula why??and over glass slab or non absorbent paper.
- Stainless steel spatula can react chemically with the cement.
- The powder is divided into two equal portions.
- The first portion of powder is mixed with a stiff spatula into the liquid before the next portion.

Glass Ionomer Cement

Manipulation:

- The mixing time should be 30 to 60 seconds, depending on the product.
- The cement is used immediately as the working time after mixing is about 2 minutes at 22° C.
- The cement should be dispensed, mixed ,extended working time and used while it still has its glossy surface why?? (as Zn polycarcoxylate cement).
- The cement shouldn’t be used once a “skin” forms on the surface or when the consistency becomes thicker why???

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Manipulation:

- Capsules are mixed into amalgamator or mixer according to manufacturer instruction then the mixed cement is extruded out of the capsule by applicator.
- The mixed cement is subjected to shear stresses while it is extruded from the capsule nozzle so increasing its fluidity due to pseudoplastic nature of the cement.
- The restoration should be covered with coat or varnish why?? after its initial setting.

Glass Ionomer Cement

Properties:

1. Biological properties:

Flouride releasing:

- Flouride release makes it anticariogenic restoration.
- The released fluoride is taken by surrounding enamel and dentine making them more resistance to acid attack.

Flouride recharging ability :

- To uptake fluoride whenever subjected to fluoride rich media then releases it again.

It has a mild irritation to the pulp . So, calcium hydroxide base is required in deep cavities why??.

Glass Ionomer Cement

2. ***Consistency and film thickness:***

- About 24 microns which is similar to that of Zn Phosphate

3. Solubility:

- The GIC is very sensitive to water in first 24 h.
- Therefore, it is necessary to coat the restoration by a varnish immediately after its insertion to protect the cement from premature exposure to the saliva or from dryness.
- After 24 h, the solubility is low.

Glass Ionomer Cement

4. Strength:

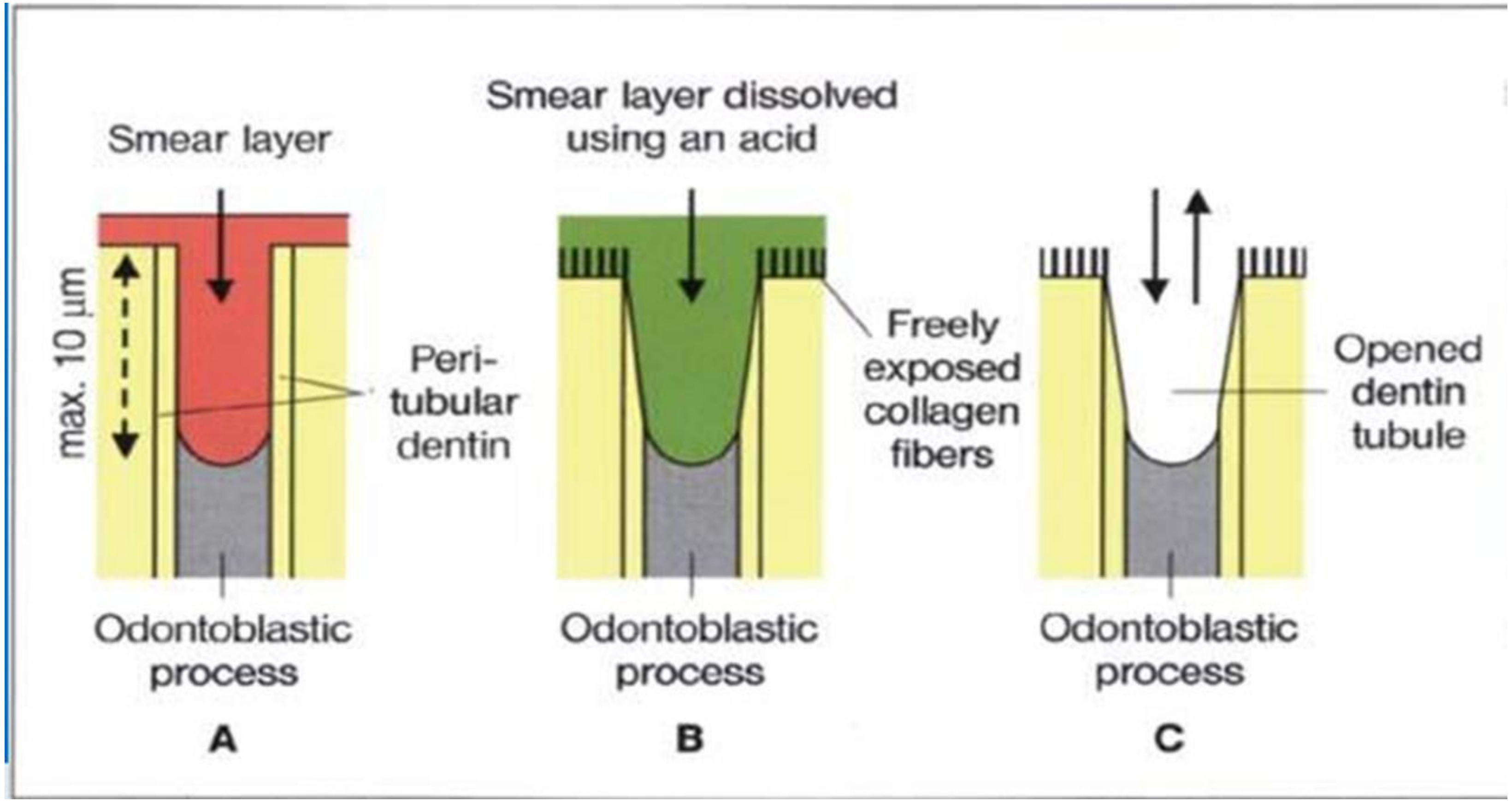
- Its mechanical properties are superior to other acid-base cements and increase over long periods.
- Compressive strength is between 100 and 150 Mpa , Strength continues for months due to gel maturation.
- Strength improves more rapidly when the cement is covered by varnish and protected form moisture during the first 24 H.
- GIC is a brittle material(its compressive strength is higher than tensile strength).
- GIC strength does not allow it to restore stress bearing area.

Glass Ionomer Cement

5. Bonding:

- GIC bonds chemically why?? to tooth structure due to reaction of its carboxylic group with calcium and phosphate cement of tooth structure.
- To obtain good bond with dentin, the surface should be treated with conditioner to add micromechanical retention (usually diluted Polycarboxylic acid) .To remove any smear layer which interferes with bonding.
- GIC bond strength is lower than Zn polycarboxylate why??due to moisture sensitivity.

Glass Ionomer Cement



Glass Ionomer Cement

5. Bonding:

Dentin surface treatment wit 10 to 20 polyacrylic acid for 5 to 20 seconds significantly increases bond strength by removing the

Smear layer

Creating microporosities for mechanical interlocking and improving the chemical interaction of polyacid with the hydroxyapatite.

Glass Ionomer Cement

Bond strength of glass ionomer to dentin and enamel varies according to commercial product, but it is around 2 to 5 Mpa in shear loading.

GIC bond well: to stainless steel, nobel and non nobel alloys and titanium

But not to high strength core ceramics like zirconia

Glass Ionomer Cement

6. Optical properties:

- It is translucent material as the unreacted core particle is glass that can transmit light.
- The translucency matches dentine translucency, so it provides reasonable matching with tooth structure.
- Used for aesthetic restoration.
- Used in non stress bearing area.

Glass Ionomer Cement Modifications

Modification to glass ionomer cement were invented to overcome its limitations such as :

1. Initial low strength.
2. Initial water sensitivity.
3. Long setting time.

1.Metal modified Glass Ionomer

<u>1. Metal-modified glass ionomer (Cermet):</u>

- This modification is mainly related to the powder.
- Fine particles of precious metals (such as silver, gold or palladium) are sintered with the glass ionomer powder particles.
- Silver is the most commonly used metal.
- The liquid has the same composition as the liquid used for the conventional GIC.

1.Metal modified Glass Ionomer

<u>**1. Metal-modified glass ionomer (Cermet):**</u>

Advantages (Compared to conventional GIC):

1. Higher abrasion resistance (resists wear).
2. Higher flexure strength .
3. Higher fracture toughness (resists crack propagation).

Drawbacks (Compared to conventional GIC):

1. More opaque because of their metal content.
2. Release less fluoride.

Uses: Cermets can be used as:

1. Core build up material.
2. Filling material for class I and II cavities in deciduous teeth.

2. Resin modified Glass Ionomer (RMGIC)

- Moisture sensitivity and a low early strength of conventional glass ionomer cements are the result of the slow nature of the acid-base setting reaction.

To overcome this

- Polymerizable functional groups were incorporated into the cement to impart more rapid curing when activated by light or chemicals. Thus, these materials set by a polymerization reaction & the acid-base reaction that continues long after polymerization.

There are two types of resin-modified glass ionomer cements: •

1-Dual cured resin modified GI.

2-Triple cured resin modified GI.

2. Resin modified Glass Ionomer (RMGIC)

- RMGI does not show the early sensitivity to moisture conditions presented by conventional glass ionomers Why?? because of the presence of a polymeric phase which prevents the loss of water and metal ions from the poly-salt matrix.

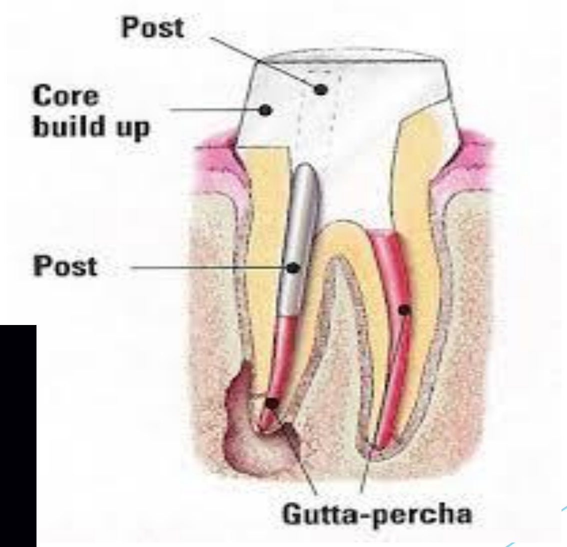
Presentation :

1. powder-liquid
2. Capsules
3. paste-paste hand-mix and auto-mix systems

2. Resin modified Glass Ionomer (RMGIC)

- Resin-modified glass ionomer luting cements are indicated for permanent cementation of:

1. Ceramic-metal crowns and fixed prostheses (e.g., metal inlays, onlays, and crowns).
2. Prefabricated or cast posts.
3. Porcelain fused to metal bridges.
4. Luting of orthodontic appliances.



2. Resin modified Glass Ionomer (RMGIC)

A. Dual cured RMGIC

- The material set by two mechanisms:

1. Acid-base reaction (between glass particles and polyacrylic acid).

2. Polymerization by light or chemical between resin components.

2. Resin modified Glass Ionomer (RMGIC)

B. Triple cured RMGIC.

The material set by three mechanisms:

1. Acid-base reaction (between glass particles and polyacrylic acid).

2. Light cured polymerization reaction of resin components.

3. Chemical-cured polymerization reaction of resin components.

- The chemical curing ensures effective polymerization in deep cavities.

2. Resin modified Glass Ionomer (RMGIC)

Two issues about HEMA must be known

2. Resin modified Glass Ionomer (RMGIC)

RMGI contains HEMA, shouldn’t be used in All ceramic restorations' cementation why??

1.HEMA is hydrophilic so the set material can uptake water by 8% leads to expansion, fracture of all ceramic.

2. Potential for color change over time and water solubility.

3- Some manufacturers do not recommend its use for luting low-strength, silica-based ceramic crowns because of the risk of fracture caused by swelling.

2. Resin modified Glass Ionomer (RMGIC)

RMGI is considered less biocompatible than conventional glass ionomers WHY???

- because of the presence of HEMA. CAUSES allergenic effect, and harmful reactions in the pulp. The largest amounts of HEMA are released in the first 24 hours, and release continues for several days.
- The release of free HEMA from the cement is higher in under-cured materials. SO correct mixing technique must be used to avoid under-curing. And In light-cured materials, the recommended exposure to light must be followed.
- Avoid using RMGI without gloves as it may cause irritation to your skin.

2. Resin modified Glass Ionomer (RMGIC)

BONDING of RMGI:

1-Bond strength to metal alloy is increased with the use of metal primers.

2-Because of the presence of methacrylate groups, these cements bond well to resin composites.

3- RMGI cements show shear bond strength to conditioned dentin or enamel in the range of 8 to 12 Mpa.

Film thickness:

Film thickness may vary between 9 and 25 µm.

3.Nano-Ionomer

- It is a resin modified glass ionomer cement with zirconia silica nano-clusters and nano-particles as additional fillers.
- It shows comparable properties to RMGIC.

4. Zirconia modified Glass Ionomer

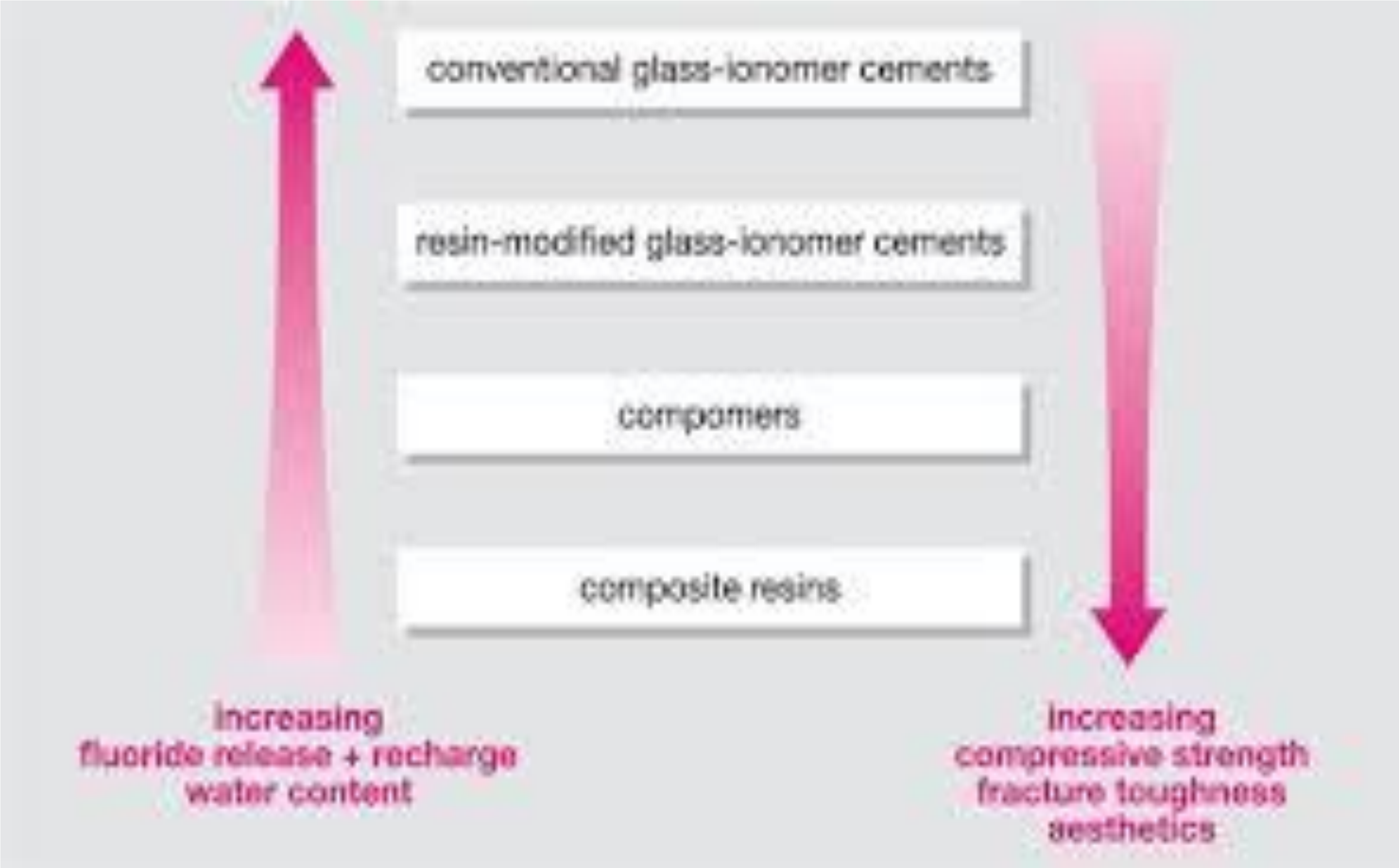
- Zirconia particles are added to conventional glass ionomer cement.
- Zirconia characterized by good mechanical properties, toughness , biocompatibility and good aesthetics.

5. Polyacid modified Resin Composite(compomer)

- The modification is more related to composite resin rather than GIC.
- Compomer is composed of methacrylate matrix modified with carboxylic groups and fillers of fluoro alumino silicate glass particles.
- The setting reaction is polymerization reaction.
- It releases fluoride but less than GIC with no recharging ability.
- It has better optical properties , less water sensitivity and superior mechanical properties compared to GIC.

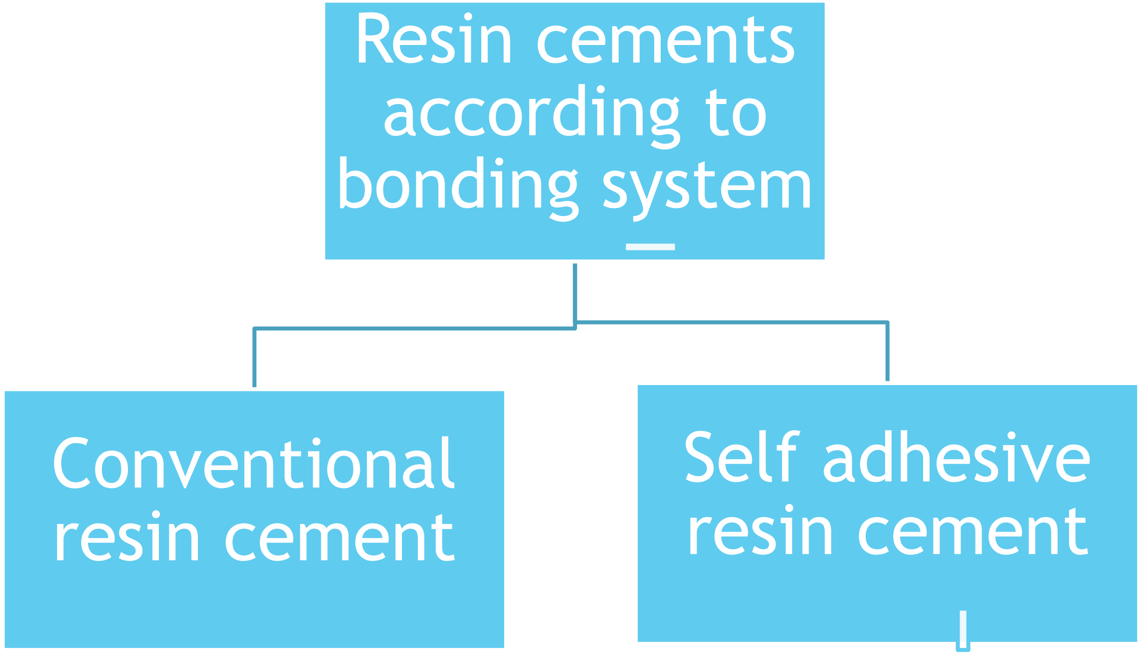
6. Giomer

- It is composed of a resin matrix and fillers.
- The fillers are fluoro alumino silicate and polyacrylic acid . Then dried, milled, silanated and incorporated into the resin matrix.


conventional glass-ionomer cements
resin-modified glass-ionomer cements
compomers
composite resins
increasing
fluoride release + recharge
water content
increasing
compressive strength
fracture toughness
aesthetics

Resin Cements

- Historically, unfilled acrylic resin cements were available but later, they became replaced by filled resin cements (resin composite cements) due to their superior properties.



Resin Cements

<u>1. Conventional resin composite cement:</u>

- They are composed of resinous matrix, fillers, initiator activator system.
- They bond with micromechanical interlocking to both restoration and tooth structure. Therefore, acid etching and bonding agents are required for cementing the restorations.

Resin Cements

Conventional resin composite cements may be further classified into three types according to the curing mechanism:

A. Self cured
B. Light cured
C. Dual cured

Resin Cements

I-Self-cured resin composite cements:

- powder/liquid or two paste systems.
- Diacrylate oligomer diluted with lower molecular weight dimethacrylate monomer is a major component .
- The other major component is silanated silica or glass filler.
- The reaction is initiated by peroxide-amine initiator accelerator system.

Resin Cements

II- Light-cured resin composite cements:

- A paste which requires no mixing.
- One major component of the cement is Bis-GMA or urethane dimethacrylate diluted with lower molecular weight dimethacrylate monomer.
- The other major component is fumed silica or glass fillers (20%-75% by weight) or both.
- A camphorquinone-amine photo-initiator system is included.

Resin Cements

III- Dual-cure resin composite cements:

- Their composition is similar to the light-cured type .
- They polymerize by the two mechanisms of setting, the light and the chemical curing at the same time.
- They are supplied as base-catalyst paste and must be mixed before use.
- After mixing, a self-cure polymerization reaction takes place slowly and provides extended working time until the cement is exposed to light , then the cement sets rapidly by a photopolymerization reaction.

Resin Cements

<u>2- Self Adhesive resin cements:</u>

- These are resin cements formulated with an adhesion promoter:
- For example adhesion promoter called (4 META).
- Another available adhesive resin cements systems achieves chemical bonding to the tooth structure via phosphonate groups.
- The phosphate end of the phosphonate reacts with the calcium of the tooth or with the metal oxide layer on the restoration surface.
- The phosphonate is very sensitive to oxygen which inhibits its polymerization, so a gel called “oxyguard” is provided to coat the margins of the restoration until the setting of the cement is completed.

Resin Cements

2- Self Adhesive resin cements:

- Self-adhesive resin cement is a class of resin-based cements that incorporates the etching, priming, and bonding chemistry in a single material.
- No need for separate etching and bonding steps, The self-adhesive resin cements were designed specifically to interact with the dentin substrate with minimal additional surface preparation.
- However, the bond to enamel is not as strong as with the use of phosphoric acid etchants. If many enamel margins are present, it is often recommended to etch with phosphoric acid even while using these cements.
- Self adhesive resin cements show good bond strength values to metal alloys and high-strength ceramics.

Resin Cements

Manipulation:

- Choose the proper shade of the resin cement.
- Isolate and dry the tooth well.
- Prepare surfaces of the tooth and the restoration (according to cement type).
- Mix the cement according to mode of supply (with plastic spatula over paper pad, automixing tip or capsule).
- Apply the cement at the internal surface of the restoration.

Resin Cements

- Apply light (for light cure and dual cured cements) for 3 seconds then remove the excess cement. Then complete the light application.
- Restorations thicker than 2.5mm should be cemented by chemical cured resin cements.

Resin Cements

Properties:

1. biological properties:

- Unreacted monomers cause irritation to the pulp. Therefore, liner should be used if remaining dentine thickness is less than 0.5 mm.
- Dual cure cements are less irritant.

Resin Cements

<u>Properties</u>

2. Consistency and film thickness:

- They are varying according to the product.

3. Solubility:

- Insoluble in oral cavity.

4. Strength:

- It depends on filler content and degree of conversion.

Resin cements have better mechanical properties compared to acid base cements.

Resin Cements

5. Bonding:

Differs according to cement type:

- Conventional resin cement bonds by micromechanical interlocking via acid etching technique,
- bonding of adhesive types occurs chemically by the help of 4-META or phosphonate

6.Optical properties:

- Translucent
- Supplied with different shades

Cements for pulp protection

- Cavity varnish
- Calcium hydroxide liner

Cavity Varnish

- It is composed of resin dissolved in organic solvent (choloroform or acetone).
- It is painted over the tooth surface using brush.
- After painting the solvent evaporates leaving non intact resin layer. Therefore 2-3 layers are applied to ensure formation of a uniform layer.

Cavity Varnish

- **Uses:**

1. Seal dentinal tubules against chemical irritants as zinc phosphate cement.
2. Seal dentinal tubules against dental amalgam corrosion products.
3. Protect glass ionomer restoration from early moisture sensitivity.

Calcium Hydroxide

- **<u>Mode of supply:</u>**

1. Two paste in two collapsible tubes(base and catalyst).
2. Light cured containing resin matrix and photo activation system.

Calcium Hydroxide

- **Manipulation:**

1. Two pastes: equal volumes of two pastes are mixed for 30 seconds and set in about 2 minutes.
2. Light activated: dispensed inside the cavity and cured for 40 seconds.

Calcium Hydroxide

- **Properties:**

1. Biological properties:

- The freshly mixed cement is highly alkaline (PH: 11-12).
- This alkalinity results stimulation of the pulp and formation of secondary dentine when placed in very deep cavities (indirect pulp capping) or directly on pulp (direct pulp capping).
- Its high alkalinity results in bactericidal effect.

Calcium Hydroxide

2. Solubility:

- It is highly soluble.

3. Strength:

- It has very low mechanical properties.

In deep cavities :

- Place thin layer of $Ca(OH)_2$ as subliner.
- Then zinc phosphate as base.
- Never to place amalgam directly on $Ca(OH)_2$ Why??▪ Can’t withstand force of amalgam condensation ▪ Will not provide adequate thermal insulation

Calcium Hydroxide

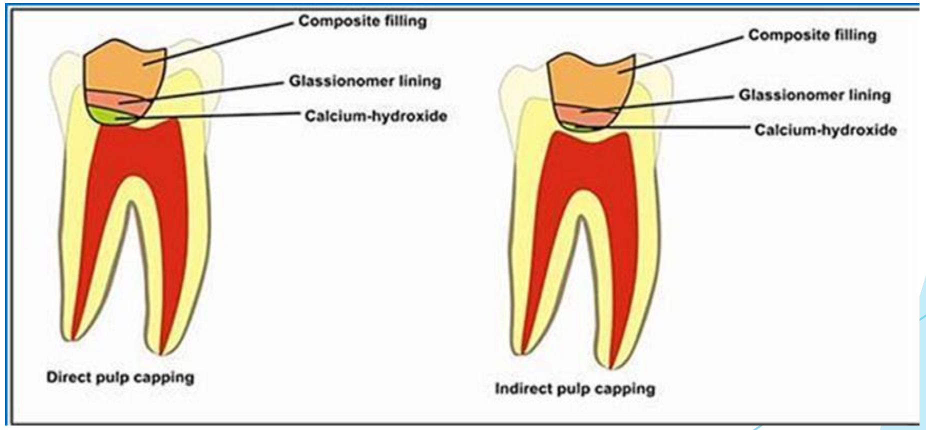
4. Bonding

- Micromechanical interlocking.

5. Optical properties

- Opaque so it cannot be used under esthetic restorations.

Figure5. Glass ionomer cement used was a liner (sandwich technique)




THANK YOU!